St. George's University
Grenada, West Indies

Basic Principles of Medicine 2

DM: Digestive System & Metabolism | Lecture 11

Clinical Gastrointestinal System – Flipped Classroom

Dr. Stephen Onigbinde

sonigbin@sgu.edu

Department of Anatomical Sciences

School of Medicine, St. George's University

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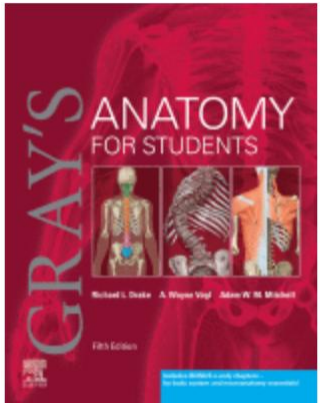
Objectives

SOM.MK.1.BMP2.3.DM.1.ANAT.1124	Explain where free air and fluid in the peritoneal cavity usually accumulates.
SOM.MK.1.BMP2.3.DM.1.ANAT.1128	Describe achalasia and its clinical presentation
SOM.MK.1.BMP2.3.DM.1.ANAT.1133	Compare and contrast where stomach contents from a perforation of the posterior, versus the anterior stomach wall will initially accumulate.
SOM.MK.1.BMP2.3.DM.1.ANAT.1134	List the blood vessels which may be eroded in ulceration of the posterior wall of the stomach.
SOM.MK.1.BMP2.3.DM.1.ANAT.1139	Identify and describe the large artery at risk for injury/hemorrhage by a perforated ulcer in the posterior wall of the duodenal bulb (1st part of duodenum).
SOM.MK.1.BMP2.3.DM.1.ANAT.1140	Explain the passage tumor cells may take via the lymphatic drainage from different portions of the small large intestine
SOM.MK.1.BMP2.3.DM.1.ANAT.1143	*Describe the anatomical location, mechanism and clinical sequences of renal vein entrapment syndrome (review CPR)*
SOM.MK.1.BMP2.3.DM.1.ANAT.1144	Describe the psoas sign and explain why it can be positive during appendicitis.
SOM.MK.1.BMP2.3.DM.1.ANAT.1145	Explain the surgical significance of Mc Burney's point.
SOM.MK.1.BMP2.3.DM.1.ANAT.1170	List the 3 common sites where gall stones can be impacted (lodged) and explain the effects of each on the bile flow.
SOM.MK.1.BMP2.3.DM.1.ANAT.1171	Explain how the visceral pain from an inflamed gallbladder can refer to the shoulder.
SOM.MK.1.BMP2.3.DM.1.ANAT.1180	Explain the pathway taken by an endoscope from mouth to the biliary tree during Endoscopic Retrograde Cholangio Pancreatography (ERCP) to remove a gallstone
SOM.MK.1.BMP2.3.DM.1.ANAT.1181	Describe the portal circulation and the areas of anastomosis with the systemic circulation
SOM.MK.1.BMP2.3.DM.1.ANAT.1182	Describe the anatomical basis for the clinical symptoms of portal hypertension including esophageal varices, rectal varices and caput medusa
SOM.MK.1.BMP2.3.DM.1.ANAT.1183	Explain how portal hypertension can be relieved by portacaval shunts

Recommended Reading

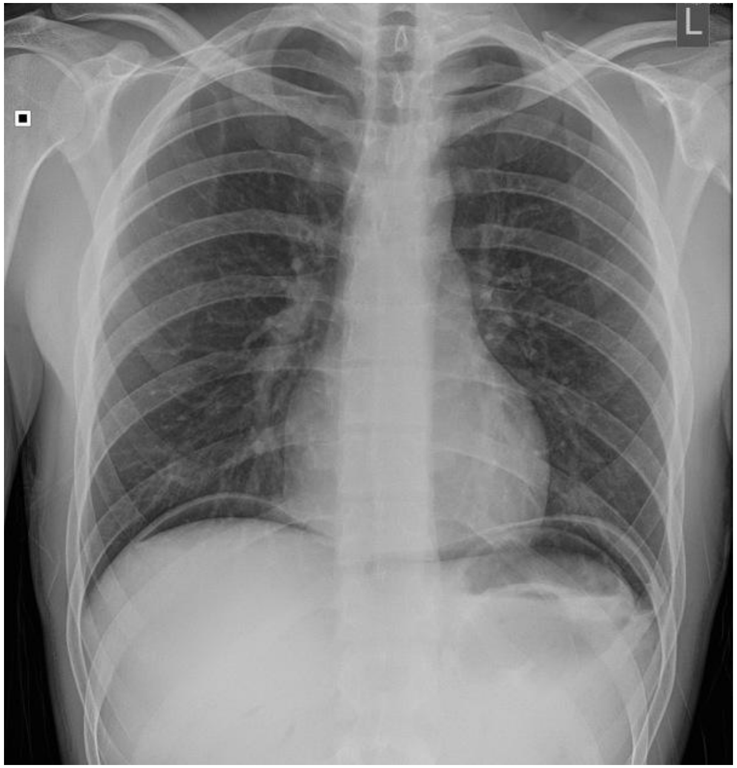
- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Online Textbook: Chapter 4: Click here
    - Review this packet and address case questions
    - Previous content [DLA & , DM Lecture 3, 5, 7, 9]
    - In the Clinic: Duodenal ulceration
    - In the Clinic: Appendicitis
    - In the Clinic: Gallstones
    - In the Clinic: ERCP
    - In the Clinic: Portosystemic Anastomosis
- Pay special attention to the green (print text)/blue (ebook)“In the Clinic” boxes for the clinical correlations for this topic



Case 1

A 44-year-old man comes to the emergency department because of acute onset abdominal pain, nausea, vomiting and generalized weakness. Pain is localized to the epigastric region with radiation to the back and is described as a burning sensation. Patient has a history of smoking and heavy alcohol use. Blood pressure is 112/90mmHg, pulse is 120/min, respirations are 26/min and temperature is 38.4^{o}C (101^{o}F). Physical exam shows a distressed patient with a rigid abdomen and epigastric tenderness.

Labs show increased white blood cells. X-ray is shown. Following stabilization, CT confirms radiographic findings and shows free fluid all over the abdominal cavity. During exploratory laparotomy, purulent fluid with undigested products was seen in all four quadrants. After removal of fluid, a perforation in the first part of the duodenum, just distal to pylorus was seen. Surgery was completed with no post-op complications.



Case Questions

1. Which blood vessel is most at risk for injury by a perforated ulcer in the posterior wall of the duodenal bulb?

2. Which blood vessel(s) will most likely be eroded on ulceration of the posterior wall of the stomach?

3. Where would the stomach contents from a perforation of the following areas most likely accumulate first?
   a) Posterior stomach wall
   b) Anterior stomach wall
   c) Which most likely occurred in the patient in Case 1 (assuming no changes in position)?

4. Where will free air most likely accumulate in the peritoneal cavity?

5. Describe the pathway of fluid around the abdominal cavity.

Case 2

A 65-year-old woman comes to the urgent care because of abdominal pain, nausea, loss of appetite and mild fever for the last 24 hours. The pain was initially dull and periumbilical but is now in the right lower quadrant. Blood pressure is 126/86mmHg, pulse is 68/min, respirations are 16/min and temperature is 36°C (96.8°F). Physical examination shows rebound tenderness in the right lower quadrant (RLQ) and a positive psoas sign. CT shows a metal-density foreign body in the right lower quadrant. Patient is referred for surgical treatment of suspected appendicitis.

1. Explain the reason for the change in the patient's pain complaint.

2. Explain the following signs and why they can be positive during appendicitis:
   a. Psoas sign
   b. Rebound tenderness

3. Describe Mc Burney's point and explain its significance during an appendectomy.

Case 3

A 45-year-old man comes to the emergency department because of two episodes of bloody vomiting over the past week. Patient also complains of dizziness and melena (dark, tarry stool). Patient has a history of chronic alcohol use. His blood pressure is 138/88mmHg, pulse is 104/min, respirations are 16/min and temperature is 38.9^{o}C (102^{o}F). Lab tests show low hemoglobin (anemia). Endoscopy confirms esophageal varices. CT scan shows enlargement of the liver and spleen, as well as dilated and tortuous splenic and portal veins. After additional testing, a diagnosis of portal hypertension is confirmed.

1. Define hematemesis and hematochezia.

2. Describe the portal circulation and the areas of anastomosis with the systemic circulation.

3. Describe portal hypertension and its relationship to esophageal and rectal varices, and caput medusa.

4. Describe how portal hypertension can be reduced.

Case 4

A 73-year-old woman comes to the emergency department because of nausea and vomiting after meals for the past 2 days. Patient also complains of loss of appetite, bloating and mild epigastric discomfort and right shoulder pain. Her blood pressure is 124/78mmHg, pulse is 104/min, respirations are 16/min and temperature is 38^{o}C (100.4^{o}F). Physical exam shows yellowing of the eyes. Laboratory tests show elevated serum bilirubin. Due to symptoms and lack of findings on US, CT and inflamed gallbladder on MRCP, patient is scheduled for Endoscopic retrograde cholangiopancreatography (ERCP) to confirm and treat the suspected obstruction within the biliary tree.

1. Define and explain Endoscopic Retrograde CholangioPancreatography (ERCP).

2. Explain why visceral pain from an inflamed gallbladder can be referred to the shoulder.

3. List the 3 common sites where gall stones can be impacted and explain the effects of each on bile flow.